# National University of Computer and Emerging Sciences, Lahore Campus

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| **Course:** | **Data Mining** |
| **Program:** | **BSE, BCS** |
| **Submission Date:** | **09-04-26** |
| **Section:** | **6A, 8A** |
| **Assignment no. :** | **03** |
| **Total Marks:** | **50** |

**Important Instructions:**

- **Use A4 sheets or narrow line sheets for solving assignments.**
- **Assignments done on register pages will be cancelled.**
- **Must attach the assignment title page shared to you at the end of assignment.**
- **This assignment must be completed independently by the student.The use of AI tools (including ChatGPT, Bard, Copilot, or any other generative AI system) is strictly prohibited.**
- **Students are expected to show all manual calculations and intermediate steps clearly.**

## Question 1: Naive Bayes (10 Marks)

| Outlook | Temperature | Humidity | Wind | Play |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Overcast | Hot | High | Weak | Yes |
| Rain | Mild | High | Weak | Yes |
| Rain | Cool | Normal | Weak | Yes |
| Rain | Cool | Normal | Strong | No |
| Overcast | Cool | Normal | Strong | Yes |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Rain | Mild | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |
| Overcast | Mild | High | Strong | Yes |
| Overcast | Hot | Normal | Weak | Yes |
| Rain | Mild | High | Strong | No |

Tasks:

a) Calculate prior probabilities P(Play=Yes) and P(Play=No).

b) Compute conditional probabilities:

P(Outlook=Sunny | Play=Yes)

P(Humidity=High | Play=No)
P(Wind=Weak | Play=Yes)
c) Classify the instance:
Outlook = Sunny, Temperature = Cool, Humidity = High, Wind = Strong.
d) Explain why Naive Bayes is called naive and what happens if independence assumption is violated.

## Question 2: FP Growth Algorithm (10 Marks)

Given transaction database:

| Transactions ID | Itemsets |
|---|---|
| T1 | A,B,C |
| T2 | A,C,D,E |
| T3 | B,C,E |
| T4 | A,B,C,E |
| T5 | B,E |

Minimum Support = 2
a) Find support count of each item.
b) Order items according to support.
c) Construct FP Tree.
d) Generate conditional pattern base for item E.
e) List all frequent itemsets.

## Question 3: Rule Based Classification (10 Marks)

| Age | Income | Credit Rating | Class |
|---|---|---|---|
| Youth | High | Fair | No |
| Youth | High | Excellent | No |
| Middle | High | Fair | Yes |
| Senior | Medium | Fair | Yes |
| Senior | Low | Fair | Yes |
| Senior | Low | Excellent | No |
| Middle | Low | Excellent | Yes |
| Youth | Medium | Fair | No |
| Youth | Low | Fair | Yes |
| Senior | Medium | Excellent | Yes |

Tasks:
a) Generate at least three classification rules.

b) Calculate support and confidence of one rule.
c) Explain sequential covering algorithm.

## Question 4: Building Classification Rules (10 Marks)

| Attribute A | Attribute B | Class |
|---|---|---|
| 1 | Yes | Positive |
| 1 | No | Positive |
| 0 | Yes | Negative |
| 1 | Yes | Positive |
| 0 | No | Negative |
| 0 | Yes | Negative |
| 1 | No | Positive |

Tasks:
a) Build classification rules manually.
b) Identify rules with highest confidence.
c) Explain rule accuracy, coverage, and confidence.

## Question 5: Classification Model Evaluation (10 Marks)

Confusion Matrix:

| | Predicted Yes | Predicted No |
|---|---|---|
| Actual Yes | 120 | 30 |
| Actual No | 20 | 130 |

Tasks:
Calculate Accuracy, Precision, Recall, F1 Score, and Specificity.
Also interpret model performance.

## Question 7: (10 Marks)

a) Why is FP Growth preferred over Apriori?
b) When Naive Bayes perform better than Decision Trees?
c) Why is accuracy alone not enough?
d) What happens with imbalanced datasets?

# National University of Computer and Emerging Sciences

## Department of Computer Science, School of Computing

# DATA MINING

**Assignment # ____**

**Section: ____________________**

**Submitted by:**

**Name: _________________________________**

**Roll no: ________________________________**

**Submitted to:**

**Dr. Muhammad Aasim Qureshi**

**Date Assigned: ________________________________**

**Date Submission: ______________________________**